QP CODE: 25900298

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION October 2025

THIRD SEMESTER

Discipline Specific Elective (DSC) –MG3DSECND200

ADVANCEDNUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2Hours Maximum Marks: 70

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions

Each question carries 1 mark

1. C) - Activity cost
2. B) - Antidiuretic hormone
3. D) - Cholecystokinin
4. B) – NDP Cal%
5. A) – 3.5 mg/dl
6. A) – Vitamin D
7. Transcobalamins I, II, III
8. Myxoedematous
9. Phytates
10. Human milk

Part B
Short Answer
Answer any **5**questions.
Each question carries **2** marks.

11. Define calorimetry.

The process of measuring the amount of heat released or absorbed during a chemical reaction. It is used to determine the energy value of food and to determine the BMR.

12. State how water act as a part of structure

Water is a part of all tissues and is essential for growth – ***(1/2 Mark)***

Glycogen is 2/3rd water – ***(1/2 Mark)***

Fat tissue is 1/5th water– ***(1/2 Mark)***

Muscle is 3/4th water– ***(1/2 Mark)***

13. Discuss about gums.

- Occurrence - Secreted at the site of plant injury by specialised secretory cells and can be exuded from the plants **(1/2 mark)**

- Chemical nature - Gums are composed of variety of sugar and sugar derivatives. Gums are composed of a variety of sugars like galactose and glucuronic acid as well as secretory uronic acids, arabinose etc. Gums are highly fermented by colonic bacteria. **(1 mark)**

- Source - Gum arabic, oatmeal, barley, legumes **(1/2 mark)**

14. Discuss the haematological manifestations in pyridoxine deficiency

- Pyridoxine responsive hypochromic microcytic anaemia due to genetic failure of incorporation of iron into haemoglobin. ***(1 mark)***
- It is characterized by hyperferremia with sideroblasts in bone marrow. ***(1 mark)***

15. Discuss how vitamin A deficiency leads to infection

- The action of cilia of the epithelial cells is involved in protecting the body against infection
- In vitamin A deficient keratinised cells, the cilia are lost and the body is more vulnerable to infection.
- Both specific and non-specific protective mechanisms are impaired.
  - The humoral response to bacterial, parasitic and viral infections, cell mediated immunity, mucosal immunity, natural killer cell activity and phagocytosis are impaired. ***(1/2 Mark for each point)***

16. Describe the absorption of magnesium

- Magnesium absorption within the jejunum and ileum though passive and active transport mechanisms. ***(1/2 Mark)***
- The efficiency of absorption varies with the magnesium status of the individual, the amount of magnesium in the diet, and the composition of the diet as a whole. ***(1/2 Mark)***
- Magnesium level is regulated by the intestine, kidney, and bone. ***(1/2 Mark)***
- The kidney is the major regulator of magnesium excretion, but some magnesium is also lost via the stool. A drop in plasma magnesium levels stimulates the secretion of parathyroid hormones which in turn increases the absorption.***(1/2 Mark)***

17. Indicate the relationship between ageing and calcium absorption

- Calcium absorption is decreased during old age. Vitamin D deficiency and hyper parathyridism may increase.***(1/2 Mark)***
- Low vitamin D intake or inadequate exposure to sunlight reduces calcium absorption.***(1/2 Mark)***
- The efficiency to produce vitamin D by elderly is low.***(1/2 Mark)***
- Aging is also characterized by achlorhydria, which results in less gastric acidity and reduced calcium absorption.***(1/2 Mark)***

Part C

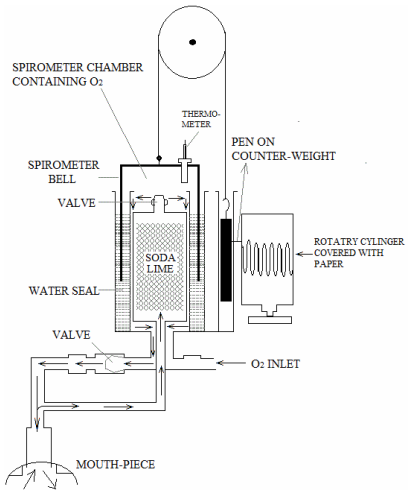
Short Essay

Answer any **6** questions

Each question carries **5** marks.

18. Explain the functioning of Benedict Roth Respirometer

Diagram – 1mark



The apparatus consists of a bell spirometer, a kymograph drum, timer, mouthpiece and its attachments. The bell spirometer is a hollow double walled cylindrical vessel. The space

between the two walls contain water. A hollow cylindrical bell fits inverted between the double walled vessel. A can containing soda lime is placed inside the double walled vessel. Kymograph drum is a revolving drum with a kymograph sheet fixed on it. It is attached to inverted vessel with the help of a pulley on one side and the other end has a pen attached to drum. A timer is attached to kymograph drum such that when timer starts, drum starts revolving at a speed in which it completes one rotation in one minute. The thermometer attached to the apparatus measures the room temperature. The mouth piece has two tubes, one supplies oxygen for inspiration and other attached to can containing soda lime through which expired air passes. The tubes are regulated by valves such that it functions in one way direction.

The subject is asked to sit or lie down quietly for the test. The spirometer is filled with oxygen. The mouthpiece is inserted to the mouth of the subject and nose is clipped. The subject is asked to breathe slowly through the mouth. When breathing is regular, connect him to apparatus by switching on oxygen supply. Then switch on the timer. As the subject breathes, spirometer bell moves up and down. As the movement of spirometer progresses, the graph is recorded on the drum. Continue recording for 8-20 minutes. From the graph the metabolic rate can be determined by calculation method. **(4 marks)**

19. Discuss the role of fibre in preventing colon cancer.

- The fibre dilutes bile acids or binds to it thereby preventing its role in mutation or cell proliferation. Fermentation of dietary fibre results in production of short chain fatty acids lowering the intestinal pH. This inhibits conversion of primary bile acids to secondary bile acids.
- The secondary bile acids are believed to promote mutation in intestine. At low pH the solubility of free bile acids is reduced, diminishing their availability for carcinogenic activity. Fermentation of dietary fibre results in production of butyrate which has been shown to be antineoplastic.
- Dietary fibre exerts its beneficial effect by speeding the passage of faeces through the large intestine so that carcinogens are in contact with the intestinal wall for much shorter period of time. Additionally the bulk and water of the faeces may dilute the carcinogen to a nontoxic level.

- Dietary fibre also influences the colonic microbial metabolism, influences fermentation in the colon and the production and distribution of short chain fatty acids in the colon. It modifies pH, increases the faecal nitrogen and influences mutagens and faecal enzymes in the colon.
- Fructo Oligo Saccharides have both complementary and overlapping mechanism of action, including the induction of detoxification enzymes, inhibition of nitrosamine formation, provision of substrate for formation of antineoplastic agents, dilution and binding of carcinogens in the digestive tract, alteration of hormone metabolism and antioxidant effects. It appears extremely unlikely that any one substance is responsible for all the associations seen.

(1 mark for each point)

20. Explain the digestion, absorption and transport of protein.

- Proteolysis takes place in the gastro-intestinal tract by proteases secreted in gastric juice, pancreatic juice and those present in intestinal mucosa.
- Gastric juice is acidic i.e. the pH is 1.5—2.5. Acidic pH of the stomach has an antiseptic action that kills the bacteria and other microorganisms.
- When the proteins enter the stomach they stimulate the secretion of the hormone called **gastrin** which in turn stimulates the secretion of HCl by parietal cells of the stomach and pepsinogen from the chief cells.
- In presence of **HCl, pepsinogen** is converted to pepsin by autocatalysis resulting in removal of some of the amino acids from the amino terminal end.
- **Pepsin** is an endopeptidase and it can hydrolyse peptide bonds in the interior of a protein molecule. It hydrolyses mainly peptide bonds containing tyrosine, phenylalanine, tryptophan leucine and acidic amino acids. Pepsin hydrolyses proteins into a mixture of polypeptides.
- As the food passes from the stomach to small intestine the low pH of the food triggers the secretion of the hormone secretin into the blood. It stimulates the pancreas to secrete HCO_3 into the small intestine in order to neutralize HCl. The secretion of HCO_3 into the intestine abruptly raises the pH from 2.5 to 7.0.
- The entry of amino acids into the duodenum releases the hormone **cholecystokinin** which in turn triggers exocrine cells of the pancreas to release pancreatic juice that

contains many pancreatic enzymes like trypsinogen, chymotrypsinogen and procarboxypeptidases

- **Trypsinogen** in the small intestine is activated to trypsin by enterokinase secreted by the intestinal cells. Trypsin is an endopeptidase which acts on the peptide bonds contributed by the basic amino acids like arginine, histidine and lysine.
- **Chymotrypsin** is secreted in an inactive from called chymotrypsinogen which is activated by trypsin. Chymotrypsin is an endopeptidase specific to aromatic amino acids i.e. phenylalanine, tyrosine, tryptophan.
- **Carboxypeptidase** secreted as procarboxypeptidase is activated again by trypsin. It is an exopeptidase that cleaves the amino acids from the carboxy terminal end. Amino peptidase secreted as pro-aminopeptidase is once again activated by trypsin.
- **Dipeptidases** acts only on dipeptides and hydrolyses it into two amino acids. Rennin is an enzyme that is present in infants to help break down milk protein. **(4 marks)**

ABSORPTION AND TRANSPORT

- Protein absorption takes place in the jejunum and ileum portions of the small intestine.
- Adenosine triphosphate is the energy source the body utilizes during protein absorption. The body uses the carrier protein transport system to absorb amino acids.
- Each amino acid group has a carrier protein that is responsible for transporting it from the intestines to the mucosa cells.
- Sodium and potassium are minerals needed for the amino acids to pass from the intestines through the villi and into the bloodstream.

(1 marks – ¼ mark for each point)

21. Discuss the clinical features of pellagra

The typical clinical features are loss of weight and increased debility. The deficiency disease is Pellagra, called as the disease of 3 Ds: Dermatitis, Diarrhoea and Dementia leading to death, the fourth D. ***(2 Marks)***

- Dermal lesions - photo sensitive rash and are seen in exposed surface of body and are hyper pigmented. The lesions are percipated on exposure to sunlight, fire and radiant heat. On neck it is called Casal's Necklace.***(1 Mark)***

- Gastrointestinal changes: Mucous membrane is inflamed resulting in severe glossitis, stomatitis, oesophagitis and gastritis and enteritis. The inflammation can produce bloody diarrhoea.***(1 Mark)***

- Neurological manifestation: Insomnia, hallucination, delirium etc develops. The peripheral nerve may be affected with paraesthesia and decreased nerve conduction, incoordination and tremors, Spastic paraplegiaand exaggerated deep tendon reflexes are observed.***(1Mark)***

22. Explain scurvy- causes and clinical manifestations in adult

- **Scurvy-**severe form of vitamin C deficiency. ***(1/2Mark)***
- **Causes** - Faulty cooking habits and inadequate intake of fresh vegetables and fruits. ***(1/2Mark)***
- The **early symptoms** are nonspecific, including listlessness, fatigue, and weakness, shortness of breath, muscle cramps, aching bones, joints and muscles and loss of appetite.***(1Mark)***
- The clinical features of scurvy are characterised by gingivitis, arthralgia, depression, postural hypotension and a marked limitation to perform strenuous work. ***(1Mark)***
- Delayed wound healing occurs due to reduced synthesis of connective tissue. ***(1Mark)***
- Haemorrhages can cause anaemia of vitamin C. Pinpoint haemorrhages under the skin causing the reddish-blue spots.***(1Mark)***

23. Explain the causes and clinical features of thiamin deficiency

Causes

- Low dietary intake of thiamin
- Poor absorption due to some abnormality of the gastrointestinal tract
- Inability of tissues to accumulate adequate supply of thiamin
- Failure of the tissues to use available thiamin properly
- An increased requirement of thiamin
- Heavy alcohol intake
  (Any 4 points – 2 marks)

Clinical features

- **Loss of appetite** and is accompanied by vomiting. ***(1/2 mark)***
- **Decreased muscle tone** of gastrointestinal tract results in decreased gastric motility, distended colon and constipation.***(1/2 mark)***

- **Mental depression and confusion-**Deficiency causes mood changes, fear, disorderly thinking, mental depression, instability, fatigue and headache. The most acute symptom is the mental confusion that precedes to coma.***(1mark)***

- **Neurological changes**: - slow and unsteady gait, disturbance of posture, equilibrium and neuromuscular coordination.***(1 mark)***

24. Differentiate skeletal fluorosis and dental fluorosis

Skeletal Fluorosis	Dental Fluorosis
▪ Mainly affects the skeletal system ▪ Patient may develop stiffness, joint pain, and deformities of the spine. ▪ Often known as genuvalgum (knock knees) ▪ Young adults and adolescents are affected ▪ When fluoride content in water is higher than 2-16 mg/lit	▪ Mainly affects the teeth ▪ Mottling of permanent teeth occurs and is irreversible. ▪ Enamel loses lusture and become rough. ▪ Small pits may develop on teeth ▪ Commonly seen in children ▪ When fluoride content in water is higher than 3-5ppm

(Each point carries 1 Mark)

25. Explain the process of absorption and metabolism of phosphorus

- Inorganic phosphorus is readily absorbed, while organic phosphorus may require enzymatic breakdownfor efficient absorption. ***(1 mark)***
- The majority of phosphorus absorption takes place particularly in the duodenum. A significant portion of phosphorus is absorbed through passive diffusion. Some phosphorus absorption also involves active transport.***(1 mark)***
- Hormones like parathyroid hormone (PTH) and vitamin D play a crucial role in regulating the absorption and utilization of phosphorus. ***(1 mark)***
- In healthy people the rate at which phosphorus absorption and excretion is equal, maintaining total body phosphorus at a steady level. ***(1 mark)***
- Approximately 90 per cent of ingested phosphorus is excreted through urine, with the remaining 10 per cent lost directly from the gastrointestinal tract without being absorbed. If plasma phosphorus levels decline, reabsorption by the kidneys increases to compensate and if plasma phosphorus levels rise too high, no phosphorus is reabsorbed in the kidneys.***(1 mark)***

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

26. Explain TER and its components.

The energy requirement of an individual is the level of energy intake from food that will balance energy expenditure. The human body's total energy needs can be divided into three categories**:** **(1 mark)**

1 The energy required to maintain basal metabolism; the basic metabolic processes required to keep body alive and healthy. It accounts for 60 to 65% of total energy requirement. **(1 mark)**

List out Factors affecting BMR **– (3 marks)**

-body composition, age, sex, body size and surface area, nutritional status, genetics, temperature, pregnancy, smoking, sleep, pathology, pharmacological agents

2 The energy required to power physical activity and muscular movements. It accounts for 25 to 30% of energy requirement.

$$\frac{\text{Energy requirement (in 24 hrs)}}{\text{BMR}} = \text{PAL}$$

PAL Value for Different Activities

Category	**PAL value**
Sedentary activity	1.40 -1.69
Moderate activity	1.70-1.99
Heavy activity	2.00-2.40

(3 marks)

3 The energy released as a result of thermic effect of food meets 5 to 10 % of total energy requirement. It is a process of increased energy expenditure and the heat release occur between 1 to 3 hours after a meal. **(2 marks)**

27. Explain the digestion, absorption and transport of carbohydrates.

- Digestion of polysaccharides begins in the mouth with salivary amylase or ptyalin or α-amylase released during the process of chewing.
- The amylase is synthesized in the serous cells of the salivary glands. Amylase breaks starches into maltose and polysaccharides. Amylase is sensitive to pH and thus is inhibited in the acidic environment of the stomach.

- Pancreatic amylase is released from acinar cells into the small intestine in concert with other enzymes under the stimulus of secretin and cholecystokinin and continues the process of carbohydrate digestion.
- Starch is digested in the small intestine to simple components. Oligosaccharides and disaccharides are digested by specific enzymes in the microvillus membrane (brush border). These enzymes include maltase that break down maltose to two glucose molecules, sucrase which digests sucrose to fructose and glucose and lactase helps to digest lactose to galactose and glucose.
- Carbohydrates not digested in the small intestine pass into the large intestine where they are digested by colonic bacteria. This results in the release of short chain fatty acids like propionate, butyrate and acetate along with methane.
- The short chain fatty acids provide vital nutrition to colonocytes, but excess volumes induce diarrhea and abdominal cramping. **(7 marks)**

Absorption and Transport

- Carbohydrates are absorbed into the blood stream as glucose, galactose and fructose. The simple sugars enter portal circulation via the capillaries of intestinal villi and are transported to the liver. The fructose and galactose are converted to glycogen for storage.
- Glucose and Na^+ share the same transport system. The concentration of Na^+ is higher in the intestinal lumen compared to mucosal cells. Na^+ moves into the cells along its concentration gradient and simultaneously glucose is transported into the intestinal cells. This is mediated by the same carrier system. This energy is indirectly supplied by ATP
- The absorption of glucose is affected by the amount of sodium (Na) ions in the intestinal lumen. A high Na^+ concentration facilitates glucose influx into the intestinal mucosal cell, whereas a low Na^+ concentration inhibits influx. **(3 marks)**

28. Explain folate deficiency, causes, clinical features and its effect on foetal development

Causes-

- **Poor dietary intake**: due to poor intake of milk, fruits and vegetables.

- **Low absorption**: due to vitamin C deficiency, celiac disease,tropical sprue, impaired absorption in pregnancy. Vitamin B12 deficiency can also result in folate deficiency.
- **Increased losses**: it occur in haemodialysis
- **Increased requirement**: increased requirement due to growth and pregnancy.
- **Infestation and infection**: malarial infection and parasitic infestation.
- **Drugs:** anticonvulsant drugs and oral contraceptives may impair folate absorption.
- **Chronic alcoholism** also causes deficiency.

(3 Marks)

Megaloblastic Anaemia

Both vitamin B12 and folate are essential for DNA synthesis and deficiency of either or both causes a failure of DNA synthesis and disordered cell proliferation. Haemopoesis is particularly susceptible and division of cells is delayed and eventually halted. Morphological changes appear in the bone marrow leads to increase in RBC size.

(1 Mark)

Clinical features

Symptoms like weakness, tiredness, dyspnoea, sore tongue, diarrhoea, irritability and forgetfulness, anorexia, headache and palpitation are seen. There is weight loss. Glossitis is less common. Neurological problems are very rare. ***(2 Marks)***

Folate deficiency during pregnancy – Folic acid is needed for the synthesis of essential components of DNA and RNA which increases rapidly during growth, thereby, increasing the requirements.

There is decrease in folate absorption and an increase in urinary excretion during pregnancy. Folate deficiency causes damage to the developing foetus. It causes neural tube defectssuch as Spina bifidaandAnencephaly**.**Because the neural tube closes by 28 days of gestation, supplementation with folic acid prior to conception is useful.
(4 Marks)

29. Describe the absorption, transport, storage, and metabolism of iron in the human body

Absorption - Iron is absorbed in the duodenum and upper part of jejunum where it can exist in two forms - haeme and non-haeme iron. Haeme iron, is more easily absorbed than non-haeme iron. Iron in food can be in the ferric or ferrous form. At the acidic pH of the stomach, ferric iron can be converted to the more soluble ferrous form.Heme iron is

absorbed directly into the intestinal cells (enterocytes). Non-heme iron is converted to ferrous iron by duodenal cytochrome b.***(4 Marks)***

Transport- Once inside the enterocyte, iron can be stored as ferritin and is transported across the basolateral membrane into the bloodstream via ferroportin. In the bloodstream, iron is bound to transferrin. A significant portion of iron is transported to the bone marrow. Iron is also transported to liver and spleen. ***(2 Marks)***

Storage - Iron is stored in cells, primarily in the liver and spleen, as ferritin. When ferritin storage capacity is exceeded, iron can also be stored as hemosiderin, a more insoluble form. ***(2 Marks)***

Metabolism - When red blood cells are broken down, the iron is released and can be recycled for erythropoiesis or stored. Between 200 and 1500 mg of iron is stored in the body as ferritin and hemosiderin; 30% is in the liver, 30% is in the bone marrow, and the rest is in the spleen and muscles. Iron is lost from the body only through bleeding and in very small amounts through defecation, sweat, and the normal exfoliation of hair and skin. Daily iron loss is approximately 1 mg for men and slightly less for non-menstruating women. The loss of iron accompanying menstruation averages approximately 0.5 mg/day. ***(2 Marks)***